

Normal Equation

When using gradient descent ,it takes multiple iterations to converge to the global minimum to find the optimal value for θ . However, normal equation is essentially used to solve to find θ analytically in one step. θ in cost function is a set of values ,therefore, the partial derivative is taken for the whole equation and then set equal to zero in order to find every value until θ_n .

Mathematical formula

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

$h_{\theta}(x) = x_0\theta_0 + \dots x_n\theta_n = X^T\theta$,where X^T is a $m \times n$ matrix and θ is a $n \times 1$ matrix.

From matrix formula $A^T A = \text{Sum of elements of } A$.

$$J(\theta) = (1/2) \cdot (X\theta - \mathbf{y})^T \cdot (X\theta - \mathbf{y})$$

From $(A-B)^T = A^T - B^T$

$$J(\theta) = (1/2) \cdot ((X\theta)^T - \mathbf{y}^T) \cdot (X\theta - \mathbf{y})$$

$$J(\theta) = (1/2) \cdot ((X\theta)^T \cdot X\theta - (X\theta)^T \cdot \mathbf{y} - \mathbf{y}^T \cdot X\theta + \mathbf{y}^T \cdot \mathbf{y})$$

From $(A \cdot B)^T = B^T A^T$ and $A^T B = B^T A$

$$J(\theta) = (1/2) \cdot (X^T \cdot \theta^T \cdot X\theta - 2\mathbf{y}^T \cdot X\theta - \mathbf{y}^T \cdot \mathbf{y})$$

$$J(\theta) = (1/2) \cdot (\theta^T \cdot (X^T \cdot X) \cdot \theta - 2(X^T \cdot \mathbf{y})^T \cdot \theta - \mathbf{y}^T \cdot \mathbf{y})$$

From $\Delta_A B^T \cdot A = B$ And $\Delta_A A^T \cdot B \cdot A = 2 \cdot B \cdot A$

$$\Delta_{\theta} J(\theta) = (1/2) \cdot (2(X^T \cdot X) \cdot \theta - 2X^T \cdot \mathbf{y})$$

$$\Delta_{\theta}J(\theta) = X^T.X.\theta - X^T.y = 0$$

$$\theta = (X^T.X)^{-1}.X^T.y$$

Here the optimal minimum parameters of linear regression are found.

Advantages

- There is no need for iterations , which gradient descent needs.
- No need to choose α (Learning rate) ,which also gradient descent needs.

Disadvantages

- Need to compute $(X^T.X)^{-1}$ which costs N^3 times ,where N is the number of features.
- Slow for large numbers of features in comparison with gradient descent.